Stochastic block models for community detection in heterogeneous networks

Hamed Kabiri Kenari

Kharazmi University, Iran

Abstract

Heterogeneous networks have multiple types of nodes and edges. Single-layer stochastic block model, bipartite stochastic block model, and multiplex stochastic block model have been proposed as a tool for detecting community structure in networks and generating synthetic networks for use as benchmarks. There are a few methods constructed for heterogeneous networks. If we use single-layer stochastic block model on heterogeneous networks by flattering them, we will lose some valuable aspects of them, and the results are not accurate. In this paper, we introduce multilayer heterogeneous stochastic block models for detecting communities in heterogeneous networks. According to these models, we look at heterogeneous networks as multilayer networks, which means each edge type shows one layer. We define two types of multilayer heterogeneous stochastic block models, independent degree models and shared degree models. Independent degree models have one specific degree parameter for each layer, and shared degree models share one degree parameter for all layers. We show that our models outperform in both real-world and synthetic networks.

Keywords: Heterogeneous Networks, Stochastic Block Models, Community Detection

1. Introduction

Having multiple types of nodes and edges is the defining feature of heterogeneous networks. For instance, nodes can be doctors, patients, hospitals, and diseases in a healthcare network. The edges can be in the type of doctor-hospital (doctor works at the hospital), patient-doctor (patient treated by a doctor), and patient-disease (patient treated for disease). Figure 1 shows a heterogeneous network of location-based social network Yelp; there are users and locations as two types of nodes and has three types of edges. Each location node is related by proximity links to other locations, a user is connected to other users by friendship relation, and a customer's visits to a location can be interpreted by a user-location link.

Like homogenous networks, the important problem is to find groups or communities of nodes that relate to the rest of the network in similar ways. Finding these communities has many applications, including separating a heterogeneous network into more homogeneous subnetworks for later analysis. In this paper, we present multilayer formulations of the popular stochastic block model, which provides a statistical solution to the community detection problem for heterogeneous networks.

We can interpret heterogeneous networks as multilayer networks. According to this view heterogeneous network has some layers, some layers connect one type of node (unipartite layers), and others connect two different types of nodes (bipartite layers). Simple approaches to community detection in heterogeneous networks include applying standard community detection algorithms to a one-layer network. We look at a heterogeneous network without layers then a heterogeneous network becomes one layer network. In this method, we discard some helpful information about the heterogeneous network, and our result communities are inaccurate. To avoid these issues, heterogeneous versions of Girvan-Newman modularity have been proposed in (X. Liu and Wakita, 2014; Zhang and Chen, 2018; Song et al., 2015; Pramanik et al., 2017).

Email address: hamed.kenari.1987@gmail.com (Hamed Kabiri Kenari)

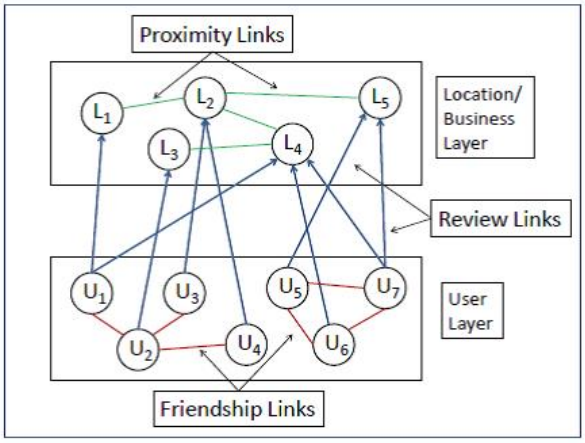

Figure 1: A sample multilayer (Yelp) network. (Pramanik et al., 2017)

One of the graceful probabilistic models for detecting communities in networks Stochastic is block models (SBMs)(Peixoto, 2013, 2014; Holland et al., 1983; Wang and Wong, 1987; Nowicki and Snijders, 2001; Karrer and Newman, 2011; Decelle et al., 2011,?; Shen et al., 2012) and for detecting communities in biological networks (**??**), and helpful in producing recommendation systems (Guimera et al., 2012). Community structure in bipartite networks can be detected by SBMs (Peixoto, 2013, 2014; Decelle et al., 2011; Shen et al., 2012; Larremore et al., 2014), and some SBM-based models have been developed for multiplex networks (Valles-Catala et al., 2016; Han et al., 2015; Peixoto, 2015; Paul and Chen, 2015; Taylor et al., 2016; Stanley et al., 2015).

In (Paul and Chen, 2022) developed several versions of multilayer stochastic models. These models are specifically for multiplex networks. It categorizes them into independent degree models and shared degree models. Also, (Paul and Chen, 2022) proposed a restricted version of MLSBMs to develop more models. In the following sections, we formulate the heterogeneous multilayer stochastic block model (HMLSBM) based on models proposed in (Paul and Chen, 2022) and describe an algorithm that searches for a maximum likelihood partition of a network into communities. We first show that planted network communities in synthetic networks can be correctly detected by the HMLSBM. We then apply the MLSBM to several real networks, showing that the MLSBM outperforms its one-layer SBM counterpart.

2. Heterogeneous Multilayer Stochastic Block Model

Our approach to the heterogeneous multilayer stochastic block models builds on recent work of (Paul and Chen, 2022), which described several versions of multilayer stochastic models. These new heterogeneous multilayer SBMs have some different versions according to degree parameters of nodes at each layers. Our new models are more effective at finding a correct partition in synthetic networks and in many real-world networks.

We begin by defining a heterogeneous network with M different types of nodes as $G = (G_U, G_B)$ where $G_U = \{L_i : i \in \{1, 2, \ldots, M\}\}$ is a set of M unipartite networks and $G_B = \{L_{ij} : i, j \in \{1, 2, \ldots, M\}, i \neq j\}$ is a set of bipartite networks that each one interconnects two different types of nodes. Each layer $L_i = (V_i, E_i)$ with V_i and E_i represent the nodes of one type and inner edges between them, respectively. Similarly, we can have layers $L_{ij} = (V_i, V_i, E_{ij})$ where $\{E_{ij} \subseteq V_i \times V_j : i, j \in \{1, 2, \ldots, M\}, i \neq j\}$ a bipartite network that connects nodes of layers L_i and L_j. We use the $N \times N$ adjacency matrix A as adjacency matrix of G that $N = \sum_{i=1}^{M} |V_i|$ and the $V_m \times V_l$ adjacency matrix $A^{m,l}$ that indicates edges of set E_{ml}, which are related as

$$A = \begin{bmatrix} A^{1,1} & \cdots & A^{1,M} \\ \vdots & \ddots & \vdots \\ A^{M,1} & \cdots & A^{M,M} \end{bmatrix}$$

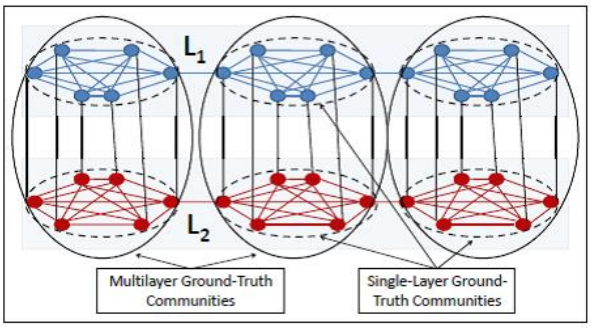

Figure 2: Network configurations with different bilayers. (Pramanik et al., 2017)

Similarly, we define community set C in a heterogeneous multilayer network G as a crossed module (C_U, C_B) of G contains a subset of single or multilayer nodes and all edges between them. Mathematically, C_U unipartite community set andC_B bipartite community set can be expressed as

$$C_U = \left\{ L_i^C = \left(V_i^C, E_i^C\right) | V_i^C \subseteq V_i, E_i^C = \left\{E_i \cap \left(V_i^C, V_i^C\right)\right\}, i \in \{1, 2, \ldots, M\} \right\}$$

$$C_B = \left\{ L_{ij}^C = \left(V_i^C, V_j^C, E_{ij}^C\right) | V_i^C \subseteq V_i, V_j^C \subseteq V_j, E_{ij}^C = \left\{E_{ij} \cap \left(V_i^C \times V_j^C\right)\right\}, i, j \in \{1, 2, \ldots, M\}, i \neq j \right\}$$

Similarly, we express the matrix of group interrelationships $\pi^{m,l}$ of layer L_{ml} as a $K \times K$ matrix where K is the number of communities in the heterogeneous network. Let $\pi_{qs}^{m,l}$ be the expected value of the adjacency matrix entry $A_{ij}^{m,l}$ for nodes i and j belonging to groups q and s ,respectively. Let the number of actual edges between i and j be drawn from a Poisson distribution with the corresponding mean $P_{ij}^{m,l}$. The communities associated with nodes i and j are z_i and z_j

$$A_{ij}^{ml} | (z_i = q, z_j = l) = A_{ji}^{lm} | (z_j = l, z_i = q) \sim Poisson(P_{ij}^{m,l})$$

The Poisson mean parameter for heterogeneous multilayer random block model can be modeled in five different ways with different number of variables. Let $\theta_i^{m,l}$control the expected degree of vertex i in network L_{ml}. The first model is an independent degree model. We write as

$$P_{ij}^{m,l} = \theta_i^{m,l} \theta_j^{m,l} \pi_{qs}^{m,l} \quad , \quad i, j \in \{1, \ldots N\}, \quad m, l \in \{1, \ldots M\} \quad q, s \in \{1, \ldots, K\}$$

With the restrictions:

$$\sum_{i:z_{i=q}} \theta_i^{m,l} = 1, \quad m, l \in \{1, \ldots, M\}, \quad q \in \{1, \ldots, K\}$$

We call it heterogeneous degree corrected multilayer stochastic Block Model (HDCMLSBM). The next model is the modified version of the DCRMLSBM, which we call it HDCRMLSBM-intra-layer. In this model we have for intra layers

$$P_{ij}^{m,l} = \theta_i^{m,l} \theta_j^{m,l} \pi_{qs} \quad , \quad i, j \in \{1, \ldots N\}, \quad m, l \in \{1, \ldots M\}, \quad m = l \quad q, s \in \{1, \ldots, K\}$$

And we have for inter layers

$$P_{ij}^{m,l} = \theta_i^{m,l} \theta_j^{m,l} \pi_{qs}^{m,l} \quad , \quad i, j \in \{1, \ldots N\}, \quad m, l \in \{1, \ldots M\} \quad m \neq l \quad q, s \in \{1, \ldots, K\}$$

With the restrictions:

$$\sum_{i:z_{i=q}} \theta_i^{m,l} = 1, \quad m, l \in \{1, \ldots, M\}, \quad q \in \{1, \ldots, K\}$$

The next model is the another modified version of the DCRMLSBM, which we call HDCRMLSBM-inter-layer. In this model we have for inter layers

$$P_{ij}^{m,l} = \theta_i^{m,l}\theta_j^{m,l}\pi_{qs} \quad , \; i,j \in \{1,\dots N\}, \; m,l \in \{1,\dots M\}, \; m \neq l \qquad q,s \in \{1,\dots,K\}$$

And we have for intra layers

$$P_{ij}^{m,l} = \theta_i^{m,l}\theta_j^{m,l}\pi_{qs}^{m,l} \quad , \; i,j \in \{1,\dots N\}, \; m,l \in \{1,\dots M\} \qquad m = l \quad q,s \in \{1,\dots,K\}$$

With the restrictions:

$$\sum_{i:z_{i=q}} \theta_i^{m,l} = 1, \qquad m,l \in \{1,\dots,M\}, \qquad q \in \{1,\dots,K\}$$

The next model is the another modified version of the DCRMLSBM, which we call HDCRMLSBM. In this model we have for all layers

$$P_{ij}^{m,l} = \theta_i^{m,l}\theta_j^{m,l}\pi_{qs} \quad , \; i,j \in \{1,\dots N\}, \; m,l \in \{1,\dots M\}, \qquad q,s \in \{1,\dots,K\}$$

With the restrictions

$$\sum_{i:z_{i=q}} \theta_i^{m,l} = 1, \qquad m,l \in \{1,\dots,M\}, \qquad q \in \{1,\dots,K\}$$

In the next model, the underlying model is shared degree, and hence the specific degree parameter of each node is shared across the layers. We call this model the heterogeneous shared Degree Restricted Multi-Layer Stochastic Block Model (SDRMLSBM). The model can be written as

$$P_{ij}^{m,l} = \theta_i^m\theta_j^l\beta_{m,l}\pi_{qs} \quad , \; i,j \in \{1,\dots N\}, \; m,l \in \{1,\dots M\}, \qquad q,s \in \{1,\dots,K\}$$

With the restrictions

$$\sum_m \sum_{i:z_i=q} \theta_i^m = 1 \;\; m \in \{1,\dots,M\}, \qquad q \in \{1,\dots,K\}$$

$$\sum_{m,l} \beta_{m,l} = 1$$

We will use the notations $e_{qs}^{m,l}$ to denote the total number of edges between communities q in layer m and community s in layer l and the degree vectors of nodes are k_i^{ml} for node i in layer m to layer l. Also, we will use the notations $e_q^{m,l}$ to denote the total number of edges in communities q in layer m to layer l, i.e., $e_{qs}^{m,l} = \sum_{i,j} A_{ij}^{m,l} I\left(z_i = q, z_j = s\right)$ and $e_q^{m,l} = \sum_{i,j} A_{ij}^{m,l} I\left(z_i = q\right) = \sum_i k_i^{m,l} I\left(z_i = q\right)$, where $I\left(.\right)$ Is the indicator function which is 1 if the condition inside is satisfied and 0 otherwise. Note that $e_{qs}^{m,l} = e_{sq}^{m,l}$, $e_q^{m,l} = \sum_s e_{qs}^{m,l}$ and $e_{qq}^{m,l}$ is twice the number of edges within the community q between layer m and l .

Similar [10, 26] we use the likelihood method, where we maximize the conditional log-likelihood $l\left(\left\{A^{m,l}\right\}|z, \left\{P^{m,l}\right\}\right)$ that obtained from the given communities and adjacent matrix. By inserting estimations of the parameter set $\left\{P^{m,l}\right\}$ conditional on z in the maximum likelihood. The conditional log-likelihood for HDCMLSBM can be written as (eliminating the terms that are not related to the communities assignment)

$$l\left(\left\{A^{m,l}\right\}; z, \left\{\pi^{m,l}\right\}, \left\{\theta^{m,l}\right\}\right) = \sum_{m,l=1}^{M} \sum_{i<j} \left\{A_{ij}^{m,l}\left\{\log\left(\pi_{z_i z_j}^{m,l}\right) + \log\left(\theta_i^{m,l}\right) + \log(\theta_j^{m,l})\right\} - \theta_i^{m,l}\theta_j^{m,l}\pi_{z_i z_j}^{m,l}\right\}$$

$$= \sum_{m,l}\sum_i k_i^{m,l}\log\left(\theta_i^{m,l}\right) + \sum_{m,l}\sum_{q\leq s}\left\{e_{qs}^{m,l}\log\left(\pi_{qs}^{m,l}\right) - \pi_{qs}^{m,l}\right\}$$

Straightforward differentiation of the log-likelihood function can estimate $\pi^{m,l}$. However we use the Lagrange multipliers to estimate $\theta_i^{m,l}$ under the constraints as follows. The objective function to be optimized is

$$l\left(\left\{\theta^{m,l}\right\}, \left\{\mu^{m,l}\right\}\right) = \sum_i \sum_{m,l} k_i^{m,l}\log\left(\theta_i^{m,l}\right) + \sum_{m,l}\sum_q \mu_q^{m,l}\left(\sum_{i:z_i=q} \theta_i^{m,l} - 1)\right)$$

Values for $\theta_i^{m,l}$ and $\pi^{m,l}$ can be estimate by solving maximize likelihood estimation as follow

$$\widehat{\theta}_i^{m,l} = \frac{k_i^{m,l}}{\sum_{i:z_i=q} k_i^{m,l}} = \frac{k_i^{m,l}}{e_q^{m,l}}$$

$$\widehat{\pi}_{qs}^{m,l} = \sum_{i,j:z_i=q,z_j=s} A_{ij}^{m,l} = e_{qs}^{m,l}$$

The maximized log-likelihood function can be obtained by replacing these estimates into the log-likelihood function as

$$l\left(\left\{A^{m,l}\right\};z\right) = \sum_i \sum_{m,l} k_i^{m,l} \log\left(\frac{k_i^{m,l}}{e_q^{m,l}}\right) + \sum_{m,l} \sum_{q\leq s} \left\{e_{qs}^{m,l} \log\left(e_{qs}^{m,l}\right) - e_{qs}^{m,l}\right\}$$

$$= \sum_{m,l} \sum_{q\leq s} e_{qs}^{m,l} \log\left(e_{qs}^{m,l}\right) - \sum_{m,l} \sum_{q\leq s} e_{qs}^{m,l} + \sum_i \sum_{m,l} k_i^{m,l} \log\left(k_i^{m,l}\right) - \sum_q \sum_{m,l} e_q^{m,l} log\left(e_q^{m,l}\right)$$

Now we drop the terms that are not related to the assignment of communities

$$l\left(\left\{A^{m,l}\right\};z\right) = \sum_{m,l} \sum_{q\leq s} e_{qs}^{m,l} \log\left(e_{qs}^{m,l}\right) - \sum_q \sum_{m,l} e_q^{m,l} log\left(e_q^{m,l}\right)$$

It is easy to write that this maximum likelihood function as follows

$$Q_{\text{HDCMLSBM}} = \sum_{m,l} \sum_{q\leq s} \left\{ e_{qs}^{m,l} \log\left(\frac{e_{qs}^{m,l}}{e_q^{m,l} e_s^{m,l}}\right) \right\}$$

Its normalized version based on each layers

$$Q_{\text{HDCMLSBM}} = \sum_{m,l} \sum_{q\leq s} \left\{ \left(e_{qs}^{m,l}/N^{m,l}\right) \log\left(\frac{\left(e_{qs}^{m,l}/N^{m,l}\right)}{\left(e_q^{m,l}/N^{m,l}\right)\left(e_s^{m,l}/N^{m,l}\right)}\right) \right\}$$

Similarly, for HDCRMLSBM-Intra-layer the condition log-likelihood with constraints can be simplified as

$$l\left(\left\{A^{m,l}\right\};z,\left\{\pi^{m,l}\right\},\left\{\theta^{m,l}\right\}\right) = \sum_{\substack{m,l=1 \\ m\neq l}}^{M} \sum_{i<j} \left\{ A_{ij}^{m,l} \left\{ \log\left(\pi_{z_i z_j}^{m,l}\right) + \log\left(\theta_i^{m,l}\right) + \log\left(\theta_j^{m,l}\right) \right\} - \theta_i^{m,l} \theta_j^{m,l} \pi_{z_i z_j}^{m,l} \right\}$$

$$+ \sum_{m=1}^{M} \sum_{i<j} \left\{ A_{ij}^{m,m} \left\{ \log\left(\pi_{z_i z_j}\right) + \log\left(\theta_i^{m,m}\right) + \log\left(\theta_j^{m,m}\right) \right\} - \theta_i^{m,m} \theta_j^{m,m} \pi_{z_i z_j} \right\}$$

$$= \sum_{\substack{m,l \\ m\neq l}} \sum_i k_i^{m,l} \log\left(\theta_i^{m,l}\right) + \sum_{\substack{m,l \\ m\neq l}} \sum_{q\leq s} \left\{ e_{qs}^{m,l} \log\left(\pi_{qs}^{m,l}\right) - \pi_{qs}^{m,l} \right\}$$

$$+ \sum_m \sum_i k_i^{m,m} \log\left(\theta_i^{m,m}\right) + \sum_m \sum_{q\leq s} \left\{ e_{qs}^{m,m} \log\left(\pi_{qs}\right) - \pi_{qs} \right\}$$

The maximize likelihood estimation of $\theta_i^{m,l}$ and π under constraints by the method of Lagrange multipliers can be written as

$$\widehat{\theta}_i^{m,l} = \frac{k_i^{m,l}}{\sum_{i:z_i=q} k_i^{m,l}} = \frac{k_i^{m,l}}{e_q^{m,l}}$$

$$\widehat{\pi}_{qs}^{m,l} = \sum_{i,j:z_i=q,z_j=s} A_{ij}^{m,l} = e_{qs}^{m,l}, \qquad m \neq l$$

$$\pi_{qs} = \sum_{m} \sum_{i,j:z_i=q,z_j=s} A_{ij}^{m,m} = \sum_{m} e_{qs}^{m,m}$$

Replacing maximize likelihood estimations into the log-likelihood equation and then eliminating terms that are not related to community assignment

$$l\left(\left\{A^{m,l}\right\};z\right) = \sum_{\substack{m,l \\ m\neq l}} \sum_{q\leq s} e_{qs}^{m,l} \log\left(e_{qs}^{m,l}\right) - \sum_{q} \sum_{\substack{m,l \\ m\neq l}} e_{q}^{m,l} log\left(e_{q}^{m,l}\right)$$

$$+ \sum_{m} \sum_{q\leq s} e_{qs}^{m,m} \log\left(\sum_{m} e_{qs}^{m,m}\right) - \sum_{q} \sum_{m} e_{q}^{m,m} log\left(e_{q}^{m,l}\right)$$

It is easy to write that this maximum likelihood function as

$$Q_{\mathrm{HDCRMLSBM-Intra-layer}} = \sum_{\substack{m,l \\ m\neq l}} \sum_{q\leq s} \left\{ e_{qs}^{m,l} \log\left(\frac{e_{qs}^{m,l}}{e_{q}^{m,l} e_{s}^{m,l}}\right)\right\} + \sum_{m} \sum_{q\leq s} \left\{ e_{qs}^{m,m} \log\left(\frac{\sum_{m} e_{qs}^{m,m}}{e_{q}^{m,l} e_{s}^{m,l}}\right)\right\}$$

Its normalized version based on each layers

$$Q_{\mathrm{HDCRMLSBM-Intra-layer}} = \sum_{\substack{m,l \\ m\neq l}} \sum_{q\leq s} \left\{ \left(e_{qs}^{m,l}/N^{m,l}\right) \log\left(\frac{\left(e_{qs}^{m,l}/N^{m,l}\right)}{\left(e_{q}^{m,l}/N^{m,l}\right)\left(e_{s}^{m,l}/N^{m,l}\right)}\right)\right\}$$

$$+ \sum_{m} \sum_{q\leq s} \left\{ \left(e_{qs}^{m,m}/N^{m,m}\right) \log\left(\frac{\sum_{m}\left(e_{qs}^{m,m}/N^{m,m}\right)}{\left(e_{q}^{m,m}/N^{m,m}\right)\left(e_{s}^{m,m}/N^{m,m}\right)}\right)\right\}$$

Similarly, for HDCRMLSBM-Inter-layer the condition log-likelihood with constraints can be simplified as

$$l\left(\left\{A^{m,l}\right\};z,\left\{\pi^{m,l}\right\},\left\{\theta^{m,l}\right\}\right) = \sum_{\substack{m,l=1 \\ m\neq l}}^{M} \sum_{i<j} \left\{ A_{ij}^{m,l} \left\{\log\left(\pi_{z_i z_j}\right) + \log\left(\theta_{i}^{m,l}\right) + \log(\theta_{j}^{m,l})\right\} - \theta_{i}^{m,l}\theta_{j}^{m,l}\pi_{z_i z_j}\right\}$$

$$+ \sum_{m=1}^{M} \sum_{i<j} \left\{ A_{ij}^{m,m} \left\{\log\left(\pi_{z_i z_j}^{m,m}\right) + \log\left(\theta_{i}^{m,m}\right) + \log(\theta_{j}^{m,m})\right\} - \theta_{i}^{m,m}\theta_{j}^{m,m}\pi_{z_i z_j}^{m,m}\right\} = \sum_{\substack{m,l \\ m\neq l}} \sum_{i} k_{i}^{m,l} \log\left(\theta_{i}^{m,l}\right)$$

$$+ \sum_{\substack{m,l \\ m\neq l}} \sum_{q\leq s} \left\{ e_{qs}^{m,l} \log\left(\pi_{qs}\right) - \pi_{qs}\right\} + \sum_{m} \sum_{i} k_{i}^{m,m} \log\left(\theta_{i}^{m,m}\right) + \sum_{m} \sum_{q\leq s} \left\{ e_{qs}^{m,m} \log\left(\pi_{qs}^{m,m}\right) - \pi_{qs}^{m,m}\right\}$$

The maximize likelihood estimation of $\theta_{i}^{m,l}$ and π under constraints by the method of Lagrange multipliers can be written as

$$\widehat{\theta}_{i}^{m,l} = \frac{k_{i}^{m,l}}{\sum_{i:z_i=q} k_{i}^{m,l}} = \frac{k_{i}^{m,l}}{e_{q}^{m,l}}$$

$$\widehat{\pi}_{qs} = \sum_{m,l} \sum_{i,j:z_i=q,z_j=s} A_{ij}^{m,l} = \sum_{\substack{m,l \\ m \neq l}} e_{qs}^{m,l}, \qquad m \neq l$$

$$\pi_{qs}^{m,m} = \sum_{i,j:z_i=q,z_j=s} A_{ij}^{m,m} = e_{qs}^{m,m}$$

Replacing maximize likelihood estimations into the log-likelihood equation and then eliminating terms that are not related to community assignment

$$l\left(\left\{A^{m,l}\right\};z\right) = \sum_{\substack{m,l \\ m \neq l}} \sum_{q \leq s} e_{qs}^{m,l} \log \left(\sum_{\substack{m,l \\ m \neq l}} e_{qs}^{m,l} \right) - \sum_{q} \sum_{\substack{m,l \\ m \neq l}} e_{q}^{m,l} log\left(e_{q}^{m,l}\right)$$

$$+ \sum_{m} \sum_{q \leq s} e_{qs}^{m,m} \log\left(e_{qs}^{m,m}\right) - \sum_{q} \sum_{m} e_{q}^{m,m} log\left(e_{q}^{m,l}\right)$$

It is easy to write that this maximum likelihood function as

$$Q_{\mathrm{HDCRMLSBM-Inter-layer}} = \sum_{\substack{m,l \\ m \neq l}} \sum_{q \leq s} \left\{ e_{qs}^{m,l} \log \left(\frac{\sum_{\substack{m,l \\ m \neq l}} e_{qs}^{m,l}}{e_{q}^{m,l} e_{s}^{m,l}} \right) \right\} + \sum_{m} \sum_{q \leq s} \left\{ e_{qs}^{m,m} \log \left(\frac{e_{qs}^{m,m}}{e_{q}^{m,m} e_{s}^{m,m}} \right) \right\}$$

Its normalized version based on each layers

$$Q_{\mathrm{HDCRMLSBM-Inter-layer}} = \sum_{\substack{m,l \\ m \neq l}} \sum_{q \leq s} \left\{ \left(e_{qs}^{m,l}/N^{m,l}\right) \log \left(\frac{\sum_{\substack{m,l \\ m \neq l}} \left(e_{qs}^{m,l}/N^{m,l}\right)}{\left(e_{q}^{m,l}/N^{m,l}\right)\left(e_{s}^{m,l}/N^{m,l}\right)} \right) \right\}$$

$$+ \sum_{m} \sum_{q \leq s} \left\{ \left(e_{qs}^{m,m}/N^{m,m}\right) \log \left(\frac{\left(e_{qs}^{m,m}/N^{m,m}\right)}{\left(e_{q}^{m,m}/N^{m,m}\right)\left(e_{s}^{m,m}/N^{m,m}\right)} \right) \right\}$$

Similarly for HDCRMLSBM the condition log-likelihood with constraints can be simplified as

$$l\left(\left\{A^{m,l}\right\};z,\pi,\left\{\theta^{m,l}\right\}\right) = \sum_{m,l=1}^{M} \sum_{i<j} \left\{ A_{ij}^{m,l} \left\{ \log\left(\pi_{z_i z_j}\right) + \log\left(\theta_{i}^{m,l}\right) + \log(\theta_{j}^{m,l}) \right\} - \theta_{i}^{m,l} \theta_{j}^{m,l} \pi_{z_i z_j} \right\}$$

The maximize likelihood estimation of $\theta_i^{m,l}$ and π under constraints by the method of Lagrange multipliers can be written as

$$\widehat{\theta}_{i}^{m,l} = \frac{k_i^{m,l}}{\sum_{i:z_i=q} k_i^{m,l}} = \frac{k_i^{m,l}}{e_q^{m,l}}$$

$$\widehat{\pi}_{qs} = \sum_{m,l} \sum_{i,j:z_i=q,z_j=s} A_{ij}^{m,l} = \sum_{m,l} e_{qs}^{m,l},$$

Replacing maximize likelihood estimations into the log-likelihood equation and then eliminating terms that are not related to community assignment

$$l\left(\left\{A^{m,l}\right\};z\right)=\sum_{m,l}\sum_{q\leq s}e_{qs}^{m,l}\log\left(\sum_{m,l}e_{qs}^{m,l}\right)-\sum_{q}\sum_{m,l}e_{q}^{m,l}log\left(e_{q}^{m,l}\right)$$

It is easy to write that this maximum likelihood function as follows

$$Q_{\text{HDCRMLSBM}}=\sum_{m,l}\sum_{q\leq s}\left\{e_{qs}^{m,l}\log\left(\frac{\sum_{m,l}e_{qs}^{m,l}}{e_{q}^{m,l}e_{s}^{m,l}}\right)\right\}$$

Its normalized version based on each layers

$$Q_{\text{HDCRMLSBM}}=\sum_{m,l}\sum_{q\leq s}\left\{\left(e_{qs}^{m,l}/N^{m,l}\right)\log\left(\frac{\sum_{m,l}\left(e_{qs}^{m,l}/N^{m,l}\right)}{\left(e_{q}^{m,l}/N^{m,l}\right)\left(e_{s}^{m,l}/N^{m,l}\right)}\right)\right\}$$

Similarly, for HSDRMLSBM, the condition log-likelihood with constraints can be simplified as

$$l\left(\left\{A^{m,l}\right\};z,\pi,\{\theta^{m}\}\right)=\sum_{m,l=1}^{M}\sum_{i<j}\left\{A_{ij}^{m,l}\left\{\log\left(\pi_{z_iz_j}\right)+log\left(\beta_{m,l}\right)+\log\left(\theta_{i}^{m}\right)+\log(\theta_{j}^{l})\right\}-\theta_{i}^{m}\theta_{j}^{l}\beta_{m,l}\pi_{z_iz_j}\right\}$$

$$=\sum_{m,l}\sum_{i}k_{i}^{m,l}\log\left(\theta_{i}^{m}\right)+\sum_{m,l}\sum_{q\leq s}e_{qs}^{m,l}\left\{\log\left(\pi_{qs}\right)+log\left(\beta_{m,l}\right)\right\}-\sum_{q\leq s}\pi_{qs}$$

The maximize likelihood estimation of θ_i^m and π and $\beta_{m,l}$under constraints by the method of Lagrange multipliers can be written as

$$\widehat{\theta}_{i}^{m}=\frac{\sum_{m,l}k_{i}^{m,l}}{\sum_{m,l}\sum_{i:z_i=q}k_{i}^{m,l}}=\frac{\sum_{m,l}k_{i}^{m,l}}{\sum_{m,l}e_{q}^{m,l}}$$

$$\widehat{\pi}_{qs}=\sum_{m,l}\sum_{i,j:z_i=q,z_j=s}A_{ij}^{m,l}=\sum_{m,l}e_{qs}^{m,l},$$

$$\widehat{\beta}_{m,l}=\frac{\sum_{q\leq s}e_{qs}^{m,l}}{\sum_{m,l}\sum_{q\leq s}e_{qs}^{m,l}}=\frac{L^{m,l}}{L}$$

It is easy to write that this maximum likelihood function as follows

$$Q_{\text{HSDRMLSBM}}=\sum_{m,l}\sum_{q\leq s}\left\{e_{qs}^{m,l}\log\left(\frac{\sum_{m,l}e_{qs}^{m,l}}{\left(\sum_{m,l}e_{q}^{m,l}\right)\left(\sum_{m,l}e_{s}^{m,l}\right)}\right)\right\}$$

Its normalized version based on each layers

$$Q_{\text{HSDRMLSBM}}=\sum_{m,l}\sum_{q\leq s}\left\{\left(e_{qs}^{m,l}/N^{m,l}\right)\log\left(\frac{\sum_{m,l}\left(e_{qs}^{m,l}/N^{m,l}\right)}{\left(\sum_{m,l}\left(e_{q}^{m,l}/N^{m,l}\right)\right)\left(\sum_{m,l}\left(e_{s}^{m,l}/N^{m,l}\right)\right)}\right)\right\}$$

3. Algorithm

We introduce an algorithm adapted by Karrer and Newman's algorithm (Karrer and Newman, 2011), for maximizing Eq. $Q_{HDCMLSBM}$, $Q_{HDCRMLSBM-Intra-layers}$, $Q_{HDCRMLSBM-Inter-layers}$, $Q_{HDCRMLSBM}$, $Q_{HSDRMLSBM}$ over all communities z, which is a modified version of the classic Kernighan-Lin algorithm (Kernighan and Lin, 1970). The algorithm gives as inputs the adjacency matrix . Then it assigns nodes at random to K communities, indexed $\{1, \ldots, K\}$ uniformly.

By choosing to move a node from one community q to another community, the algorithm searches on the functions surface to find maximum point. The algorithm chooses all the moves across all nodes and eligible communities that will most increase the functions. If there is no possible increase because moving the nodes helps get away from the local optimum, the algorithm selects the move that least decreases the functions. We move each nodes only once, and when all nodes have moved, the states through which the system has passed are evaluated and the state with the highest objective score is used as a starting point for the next search iteration. The algorithm finishes when a full iteration passes, and there is no increase in objective score.

The algorithm should be run many times because this is a kind of stochastic optimization technique, and the average score from these independent repetitions is the result of the algorithm. In these simulations, we iterate 100 times each network and an average score is reported as a result.

4. Results

In this section, we show that heterogeneous multilayer stochastic block models can detect the correct communities in synthetic networks with known planted structures and then apply them to study two real networks. For the synthetic networks, we consider nine different forms with different community structures and different layers density, which illustrate heterogeneous multilayer stochastic block models' performance under different general conditions and against alternative techniques.

On the real data sets, the first is the DBLP network; this network includes three types of nodes: paper, conference, and author. Between the three types of nodes, there are two types of edges: paper-conference (paper published at a conference) and paper-author (paper written by an author). This dataset includes 14,475 authors who wrote 14,376 papers and published them at 20 conferences. This network is commonly used as a benchmark for heterogeneous network community detection algorithms. The second is the AMiner network, which contains three types of nodes: paper, conference, and author, and two types of edges: paper-conference (paper published at a conference) and paper-author (paper written by an author). This dataset has 164,472 authors who write 127,623 papers, and published at 101 conferences.

4.1. Synthetic Networks

We examine the ability of the algorithm to extract planted structures $\pi^{m,l}$ for each layers. We describe three categories of synthetic networks, each of which illustrates a different aspect of communities in heterogeneous networks. Furthermore, each primary category has three different layers densities cases.

Each category is based on differences between the intra-layers and inter-layers of networks. In the first category, at both intra-layers and inter-layers, diagonal elements of $\pi^{m,l}$ are several times bigger than non-diagonal elements. In second category, at intra-layers diagonal elements of $\pi^{m,l}$ are several times bigger than non-diagonal elements, but at inter-layers diagonal elements of $\pi^{m,l}$ are slightly bigger than non-diagonal elements. In te third category, at intra-layers diagonal elements of $\pi^{m,l}$ are slightly bigger than non-diagonal elements, but at inter-layers diagonal elements of $\pi^{m,l}$ are several times bigger than non-diagonal elements.

Each major category has three different layers densities cases. Each case is based on differences between intra-layers densities and inter-layers densities at networks. In the first case, both intra-layers and inter-layers have equal densities. In the second case, intra-layers densities have several times bigger than inter-layers densities. In the third category, inter-layers densities have several times bigger than intra-layers densities.

To vary the number of shared communities between layers, we let the cross parameter α take values between 0 (two layers do not have common communities) and 1 (all communities are common in both layers). We construct community com_{xy} by merging the community $com_x \in C_i$ from layer L_i with the community $com_y \in C_j$ from layer L_j. Let $|C_i|$ be the number of communities of layer $L_i$and $|C_j|$ is the number of communities of layer L_j. Therefore

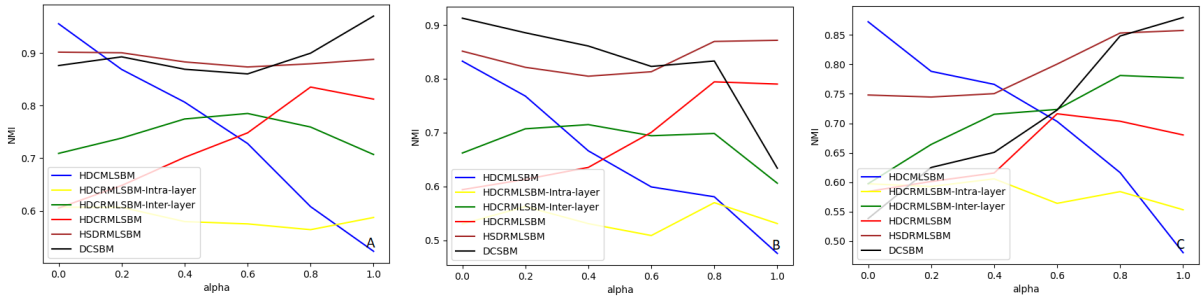

Figure 3: Simulation of strong intra-layers and strong inter-layers and α is variable (A) intra-layer-density =0.2 and inter-layer-density = 0.2 (B) intra-layer-density =0.2 and inter-layer-density = 0.05 (C) intra-layer-density =0.05 and inter-layer-density = 0.2

the maximum possible number of communities that could be combined is $|C_c| = min\{|C_i|, |C_j|\}$. We combine $|C_c| \times \alpha$ communities of both layers L_i and L_j randomly.

The normalized mutual information between correct communities and inferred communities (Danonand et al., 2005) can be a measure for accuracy. We let each community as a random variable X. Then, the normalized mutual information of the communities is $I_{norm}(X,Y) = 2I(X,Y)/[H(X)+H(Y)]$, where $H(X)$ is the Shannon entropy of X, and $I(X,Y)$ is the mutual information between correct communities and inferred communities. I_{norm} has values between 0 and 1, with $I_{norm}(X,Y) = 1$ if and only if $X = Y$, and $I_{norm}(X,Y) = 0$ when X and Y are uncorrelated.

4.1.1. First category

In this category, we define the mixing matrix as

$$\pi^{m,l} = \begin{bmatrix} \gamma & \cdots & \delta \\ \vdots & \ddots & \vdots \\ \delta & \cdots & \gamma \end{bmatrix}, \qquad \gamma = 5\delta$$

Where the variables γ, δ are positive constants. At these simulations, we produces networks with five communities. We let $M = 3$ and $N_i = 100$ for each node type and divide these vertices evenly across the five communities for each layer. To investigate the effect of density of layers on the performance of our models, we consider three different cases. Each case is based on differences between intra-layers densities and inter-layers densities at networks. In the first case, both intra-layers and inter-layers have equal densities. In the second case, intra-layers densities have several times bigger than inter-layers densities. In the third category, inter-layers densities have several times bigger than intra-layers densities. In each simulations, the cross parameter α changes between 0 (two layers don't have common communities) and 1 (all communities are common in both layers). The results are shown in Figure 3.

4.1.2. Second category

In this category, we define the mixing matrixes as

$$\pi^{m,l} = \begin{bmatrix} \gamma & \cdots & \delta \\ \vdots & \ddots & \vdots \\ \delta & \cdots & \gamma \end{bmatrix} \qquad , \qquad \gamma = 5\delta \quad , \quad m = l$$

$$\pi^{m,l} = \begin{bmatrix} \gamma & \cdots & \delta \\ \vdots & \ddots & \vdots \\ \delta & \cdots & \gamma \end{bmatrix} \qquad , \qquad \gamma = \delta + \in \qquad 0 \leq \in \leq 0.5 \quad , \quad m \neq l$$

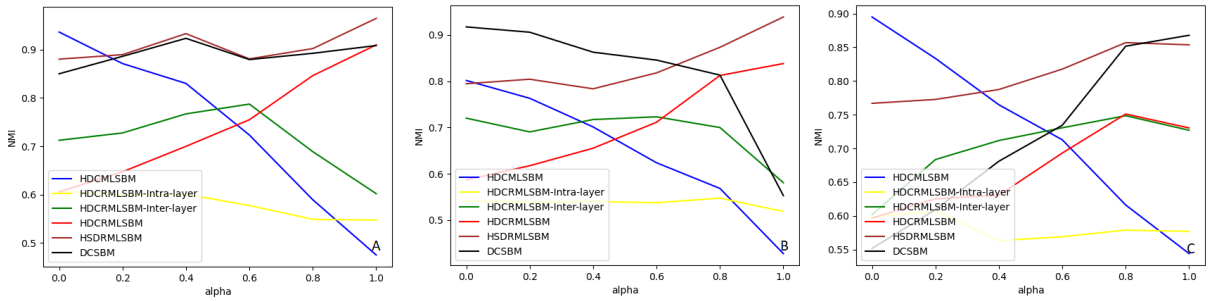

Figure 4: Simulation of strong intra layers and weak inter layers and α is variable (A) intra-layer-density =0.2 and inter-layer-density = 0.2 (B) intra-layer-density =0.2 and inter-layer-density = 0.05 (C) intra-layer-density =0.05 and inter-layer-density = 0.2

Where the variables γ, δ are positive constants. At these simulations, we produces networks with five communities. We let $M = 3$ and $N_i = 100$ for each node type and divide these vertices evenly across the five communities for each layer. To investigate the effect of density of layers on the performance of our models, we consider three different cases. Each case is based on differences between intra-layers densities and inter-layers densities at networks. In the first case, both intra-layers and inter-layers have equal densities. In the second case, intra-layers densities have several times bigger than inter-layers densities. In the third category, inter-layers densities have several times bigger than intra-layers densities. In each simulations, the cross parameter α changes between 0 (two layers don't have common communities) and 1 (all communities are common in both layers). The results are shown in Figure 4.

4.1.3. Third category

In this category, we define the mixing matrixes as

$$\pi^{m,l} = \begin{bmatrix} \gamma & \cdots & \delta \\ \vdots & \ddots & \vdots \\ \delta & \cdots & \gamma \end{bmatrix} \quad , \quad \gamma = \delta + \in \quad 0 \leq \in \leq 0.5 \quad , \; m = l$$

$$\pi^{m,l} = \begin{bmatrix} \gamma & \cdots & \delta \\ \vdots & \ddots & \vdots \\ \delta & \cdots & \gamma \end{bmatrix} \quad , \quad \gamma = 5\delta \quad , \; m \neq l$$

Where the variables γ, δ are positive constants. At these simulations, we produces networks with five communities. We let $M = 3$ and $N_i = 100$ for each node type and divide these vertices evenly across the five communities for each layer. To investigate the effect of density of layers on the performance of our models, we consider three different cases. Each case is based on differences between intra-layers densities and inter-layers densities at networks. In the first case, both intra-layers and inter-layers have equal densities. In the second case, intra-layers densities have several times bigger than inter-layers densities. In the third category, inter-layers densities have several times bigger than intra-layers densities. In each simulations, the cross parameter α changes between 0 (two layers don't have common communities) and 1 (all communities are common in both layers). The results are shown in Figure 5.

According to different simulations, the HDCMLSBM performance is better in the smaller amount of the parameter α (when layers have less common communities), and the SDRMLSBM performance is better in the larger amount of the parameter α (when layers have more common communities). It seems the reason is that when communities are less common between layers means layers communities are more independent each others therefore, HDCMLSBM that has different structures $\pi^{m,l}$ in each layers has accurate results. In contrast when communities are more common between layers means layers communities are more shared with each other; therefore, HSDRMLSBM that has same structures $\pi^{m,l}$ in each layer has accurate results.

It is exceptional in the cases where intra-layer-densities are higher, and inter-layer-densities are lower, and the parameter α is less; one layer DCSBM has better performance in comparison to HDCMLSBM. It seems the reason

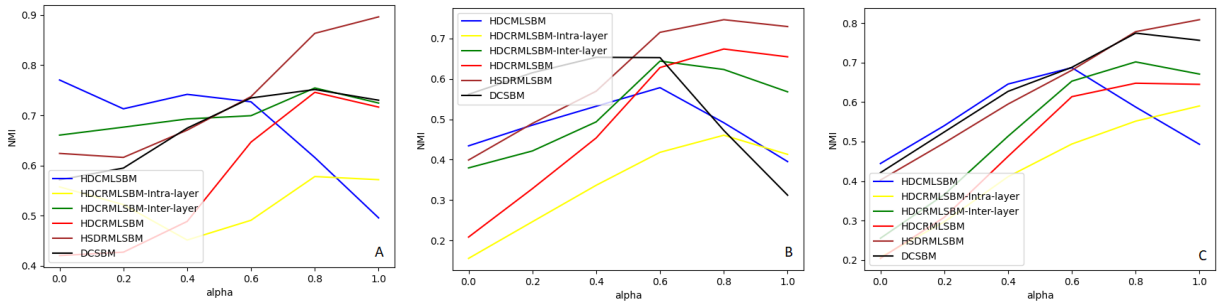

Figure 5: Simulation of weak intra layers and strong inter layers and α is variable (A) intra-layer-density =0.2 and inter-layer-density = 0.2 (B) intra-layer-density =0.2 and inter-layer-density = 0.05 (C) intra-layer-density =0.05 and inter-layer-density = 0.2

is that when inter-layers have fewer edges(less densities), connections between layers on heterogeneous are weak, therefore, detecting communities by heterogeneous multilayer SBMs are difficult than DCSBM, which does not have a multilayer view to networks.

4.2. Real Networks

4.2.1. DBLP Dataset

Our first real network is the DBLP (Digital Bibliography and Library Project) dataset (https://dblp.uni-trier.de), a common benchmark for heterogeneous networks community detection algorithms (Zhang and Chen, 2018). It is a bibliography website for computer science publications that has more than 3.4 million journal articles, conference papers, and other scientific materials. In (Gao et al., 2009) extracted a connected subset 23 of the DBLP dataset, including four research areas database, data mining, information retrieval, and artificial intelligence bibliographical records.

This network includes three types of nodes paper, conference, and author. Between the three types of nodes, there are two types of edges paper-conference (paper published at a conference), paper-author (paper written by an author). This dataset includes 14,475 authors who wrote 14,376 papers, and published at 20 conferences. Each one of the 20 conferences is labeled with the four research areas. Each research area has five conferences. 4,057 authors have true research areas that are related to a subset of 14,328 papers. The problem in this real data application is to detect the research areas of these true labeled authors correctly. Because only on labeled authors can be computed for normalized mutual information, we focus our data analysis on labeled data.

Applying our five different proposed heterogeneous multilayer SBMs and one layer DCSBM by 100 reputations. The results are shown in Table 1.

4.2.2. AMiner Dataset

Our second real network is the AMiner dataset (Tang et al., 2008) (https://www.aminer.cn/citation), which is also an academic network that includes three types of nodes, including paper, conference, and author, and two types of edges: paper-conference (paper published at a conference), paper-author (paper written by an author). This dataset has 164,472 authors who wrote 127,623 papers and published them at 101 conferences labeled with ten research areas. Just only 127,202 papers have labels for true research area. Finding the true research areas of the papers is our problem. Because just only on labeled papers can be computed for normalized mutual information, we focus our data analysis on labeled data.

Applying our five different proposed heterogeneous multilayer SBMs and one layer DCSBM by 100 reputations. The results are shown in Table 1.

5. Conclusions

In this paper, we have described multilayer stochastic block models for heterogeneous networks. We can categorize these models into two broad groups, those based on the independent degree principle and other is based on the

Table 1: The NMI of clustering from different community detection methods for DBLP and AMiner datasets

	HDCMLSBM	HDCRMLSBM-Intra-Layer	HDCRMLSBM-Inter-Layer	HDCRMLSBM	HSDRMLSBM	DCSBM
DBLP	0.920014	0.920014	0.882668	0.882668	0.885119	0.85599
AMiner	0.901216	0.901216	0.872698	0.872698	0.875139	0.84691

shared degree principle. While independent degree models for each node have different degree parameters in each layer, the shared degree base model shares degree information in layers. the independent degree model has better performance in networks that have less common communities between layers (lower parameter α). In contrast, the shared degree model has better performance in networks which have more common communities between layers (higher parameter α). But there is an exception in the case that intra-layer-densities are high and inter-layers densities are low, and the parameter α is less, one layer degree corrected stochastic block model has better performance in comparison to HDCMLSBM.

Simulation results and real data applications show the effectiveness of these proposed methods compared to single-layer methods that apply to heterogeneous networks. When we do not look at them as a multilayer view, then we don't separate node and edge types in different layers. According to our results, we cannot recommend a single model for all heterogeneous networks. We recognize some different cases which have different network elements like community structures and layers densities. The shared degree model performs better in networks that have more common communities between layers. In contrast, the independent degree model performs better in networks that have less common communities between layers.

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